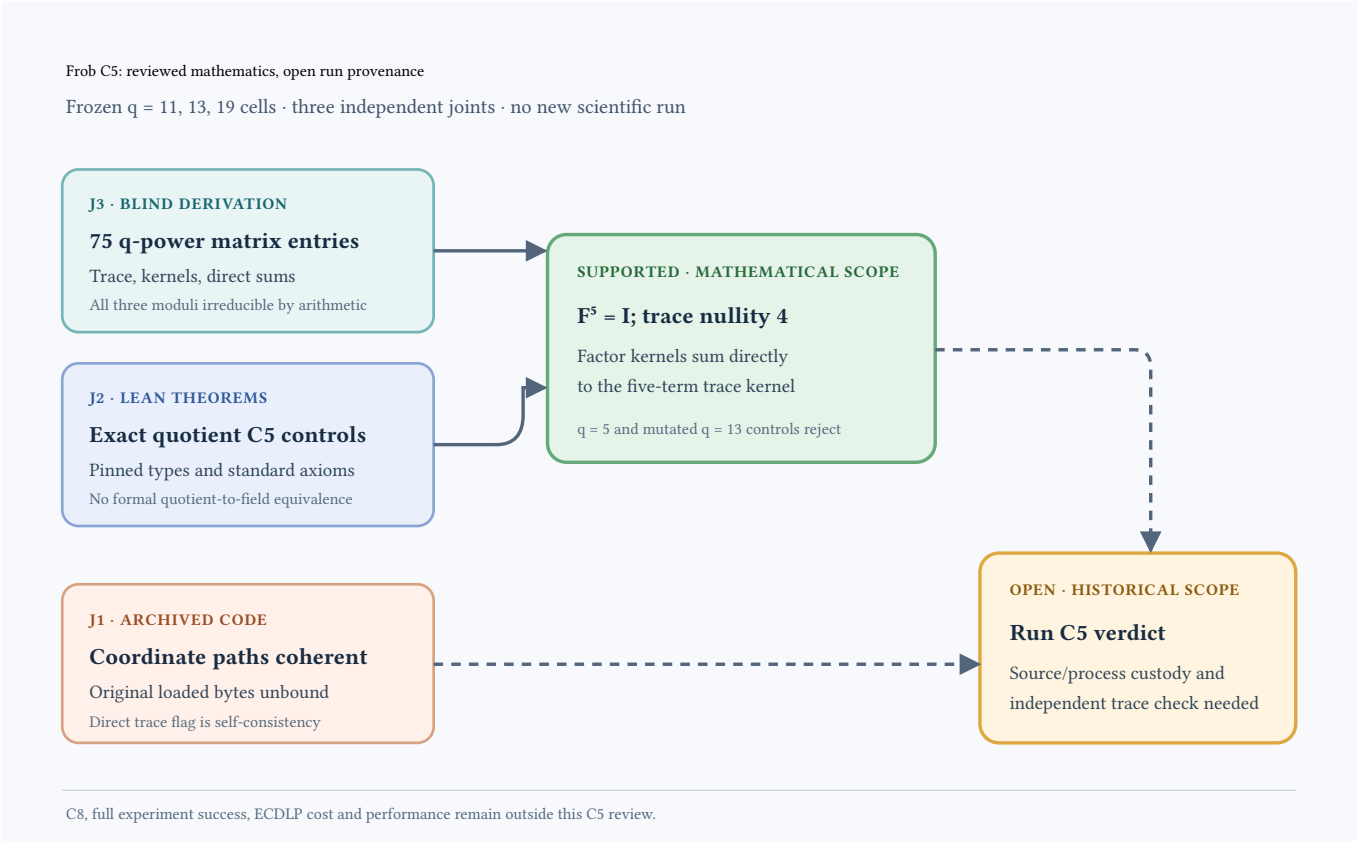
Frob C5 independent semantic review

25 October 2026 · frozen source 1c376353b8037072d17e7d98513ce3bbd1d7e993

Decision. The q = 11, 13 and 19 quotient-algebra C5 controls survive independent blind arithmetic and a scoped Lean theorem review. The historical run verdict remains **unestablished**: the frozen archive does not authenticate the exact bytes loaded by the original Sage process, and its T_equals_hM flag is not an independent direct-trace comparison.



Three independent joints

- **J1, source and code — breaks historical custody.** All 17 bound scientific files match their frozen hashes. The inspected power-basis, matrix-column and right-kernel conventions are coherent if those bytes were loaded. The original command uses a relative driver and local imports, while the manifest hashes final post-bugfix files. No original process receipt binds loaded source bytes. T and h(M) both sum powers of the same M.
- **J2, Lean — holds for mathematical scope.** The nine pinned source bodies elaborated with Lean 4.35.0-rc1 and one worker. Ten target types and 94 public producer theorem axioms were inspected. Only propext, Classical.choice and Quot.sound occur. No reviewed theorem establishes formal modulus irreducibility, a quotient-to-GaloisField equivalence, or historical executable refinement. This in-memory source-body check is narrower than a fresh per-module Lake build.
- **J3, blind arithmetic — holds for the card.** All 75 q-power entries agree with the archived replay. F to the fifth power is I in each cell; trace rank/nullity is 1/4; fixed-space dimension is 1; nontrivial kernel dimensions are 1+1+1+1, 4, and 2+2, with direct rank-four sums. The three moduli are independently irreducible by arithmetic, not by Lean theorem.

Controls and boundary

Over F_5, X^5 − 1 = (X − 1)^5, so the distinct-factor argument fails. Increasing the q = 13 matrix entry (zero-based 1,1) from 2 to 3 fails the fifth-power identity and changes trace rank from 1 to 5. The source checker independently verifies trace nullity but not the full C5 factor-kernel decomposition. C8, full experiment success, ECDLP cost and performance are outside this review.

The repository independence checker passed after the blind reviewer appended attestation metadata in a separate commit, preserving the original scientific freeze. Full source bindings, deviations, methods and report hashes are in composition.md and the archive receipt. No new Sage, EKS or scientific trial ran; no performance graph values changed.